INTERNSHIP REPORT

Occular Disease Classification using Machine Learning Models

Je Arunkumar
B M Vigneshwarlal
Deva Anand M

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Dataset Description

The dataset used for this project is the **ODIR-5K (Ocular Disease Intelligent Recognition) Dataset**, a benchmark dataset designed for multi-label classification of common ocular diseases based on retinal fundus images. It is an essential resource for developing and evaluating AI models in the field of ophthalmology.

1. Source of the Dataset

The ODIR-5K dataset was released as part of the **Grand Challenge on Ocular Disease Intelligent Recognition (ODIR)**, hosted on the **IEEE Big Data 2019** platform. It was contributed by the **Peking University International Hospital** in collaboration with other institutions, and it is publicly available through Kaggle and official challenge repositories.

2. Dataset Overview

- **Name of Dataset:** ODIR-5K (Ocular Disease Intelligent Recognition)
- **Number of Instances (Patients):** 5,000
- **Number of Fundus Images:** 7439 (2 images per patient: left and right eyes)
- **Number of Classes (Labels):** 8 disease labels + Normal
- **File Format:** JPEG for images; CSV for labels and metadata
- **Size:** ~2.0 GB

3. Features (Attributes)

The dataset includes both **image data** and **tabular metadata**. Key features are:

- **ID:** Unique patient identifier
- **Left-Fundus / Right-Fundus:** File paths to the corresponding eye images
- **Age:** Age of the patient
- **Sex:** Gender (Male/Female)
- **Left-Diagnostic Keywords / Right-Diagnostic Keywords:** Clinically annotated diagnoses
- **Label:** Binary indicators for the presence/absence of the following classes:
  - Normal (N)
  - Diabetic Retinopathy (D)
  - Glaucoma (G)
  - Cataract (C)
  - Age-Related Macular Degeneration (A)
  - Hypertensive Retinopathy (H)
  - Pathological Myopia (M)
  - Other Diseases (O)
- Number of images per class:

N	2101
D	2123
G	397
C	402
A	319
H	203

M 306
O 1588

4. Data Quality and Preprocessing

The dataset contains some variability and noise typical of medical imaging data:

- **Image Quality:** Varies across patients; some images are blurry or low contrast.
- **Missing/Uncertain Labels:** Some diagnostic keywords are incomplete or ambiguous.
- **Class Imbalance:** The dataset exhibits imbalance across disease categories, with some conditions (e.g., Diabetic Retinopathy) being more prevalent.

To prepare the data for modeling:

- Images were resized and normalized to ensure consistency.
- Labels were one-hot encoded to handle multi-label classification.
- Data augmentation techniques (e.g., flipping, rotation) were applied to increase robustness.
- Class weights were considered to address imbalance during model training.

5. Usage Context

ODIR-5K is widely used for training deep learning models to detect multiple ocular diseases from fundus images. The presence of both metadata and images makes it suitable for multi-modal learning approaches and real-world clinical decision support system development.

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METHODOLOGIES

1. ResNet (Residual Network)

Goal: Solve the **vanishing gradient problem** in very deep networks using **residual learning**.

Methodology:

1. **Residual Block:**
   - Core unit of ResNet.
   - Includes identity (skip) connections:
     `F(x) + x` where `F(x)` is the output of a few layers, and `x` is the input.
2. **Deep Architecture:**
   - Supports architectures like ResNet-18, ResNet-34, ResNet-50, etc.
   - Uses bottleneck layers in deeper versions (e.g., ResNet-50).
3. **Batch Normalization & ReLU:**
   - Applied after convolutional layers.
4. **Downsampling:**
   - Done using stride-2 convolutions instead of pooling.
5. **Global Average Pooling:**
   - Applied before the final fully connected (FC) layer.

Advantages:

- Enables training of **very deep networks** (100+ layers) efficiently.

2. VGGNet

Goal: Simplify CNN design using small (3×3) filters.

Methodology:

1. **Stacked 3x3 Convolutions:**
   - Two or three convolutions stacked before pooling.
   - Effective receptive field increases while keeping computation low.
2. **Fixed Architecture:**
   - VGG-11, VGG-16, VGG-19 – differ by number of convolutional layers.
3. **ReLU Activation:**
   - After every convolution.
4. **Max Pooling:**
   - After 2 or 3 conv layers to downsample spatial dimensions.
5. **Fully Connected Layers:**
   - 2-3 large FC layers at the end before classification.

Advantages:

- Simplicity and effectiveness.
- Used as a **feature extractor** in many applications.

3. DenseNet (Densely Connected Convolutional Network)

Goal: Improve feature reuse and gradient flow via **dense connectivity**.

Methodology:

1. **Dense Block:**
   - Each layer receives inputs from **all previous layers**:
     `x_l = H_l([x_0, x_1, ..., x_{l-1}]).`
   - `[]` denotes concatenation.
2. **Transition Layers:**
   - Used between dense blocks.
   - Contains batch normalization, 1x1 conv, and 2x2 average pooling.
3. **Growth Rate:**
   - Number of filters added per layer (e.g., 32).
4. **Feature Reuse:**
   - Encourages re-use of features, reducing redundancy.
5. **Bottleneck Layers (optional):**
   - 1x1 convs before 3x3 convs to reduce dimensions.

Advantages:

- **Fewer parameters**, better feature propagation, mitigates vanishing gradients.

4. U-Net

Goal: Efficient **semantic segmentation**, especially in biomedical imaging.

Methodology:

1. **Symmetric Encoder-Decoder Architecture:**
   - Left: encoder path (contracting) → downsampling.
   - Right: decoder path (expanding) → upsampling.
2. **Skip Connections:**
   - Connect encoder features to decoder layers of the same level.
   - Helps retain spatial information lost during downsampling.
3. **Encoder Path:**
   - Repeated Conv (3x3) + ReLU + MaxPooling.
4. **Decoder Path:**
   - Upsampling (transpose conv or upsampling) + Conv layers.
5. **Final Layer:**
   - 1x1 convolution to map feature maps to segmentation classes.

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Advantages:

- Performs well with **few training samples**.
- Preserves **context and localization** with skip connections.

WHAT IT DOES SPECIFICALLY FOR OUR PROJECT:

ResNet	Deep feature extraction; effective even with noisy/imbalanced data
VGGNet	Simple and reliable baseline; works well with transfer learning
DenseNet	Best for limited data and fine-grained classification
U-Net	Disease segmentation and localization support

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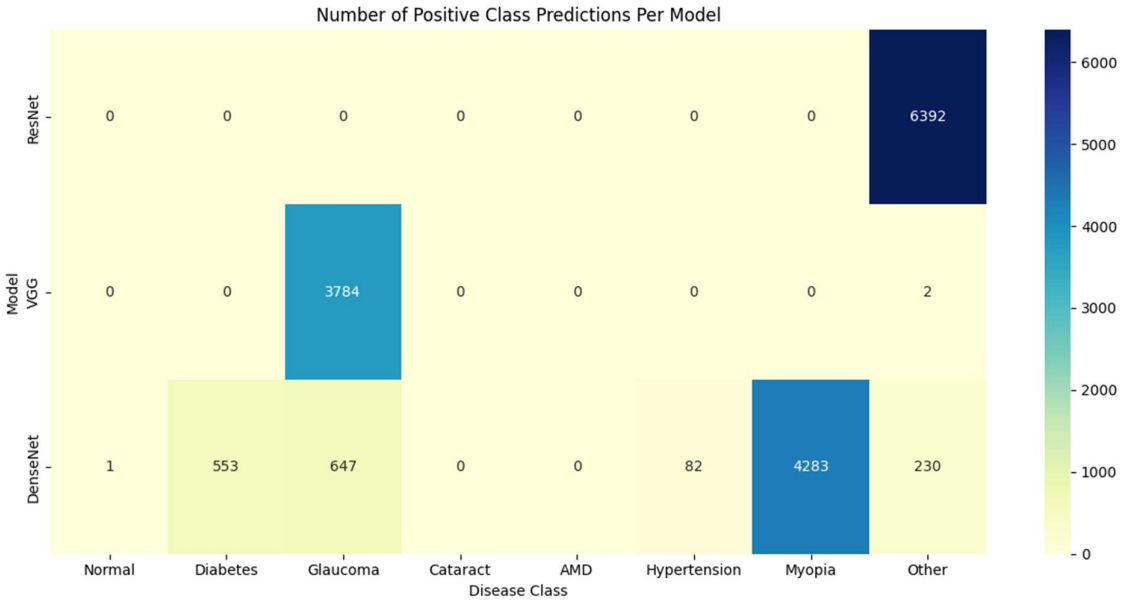
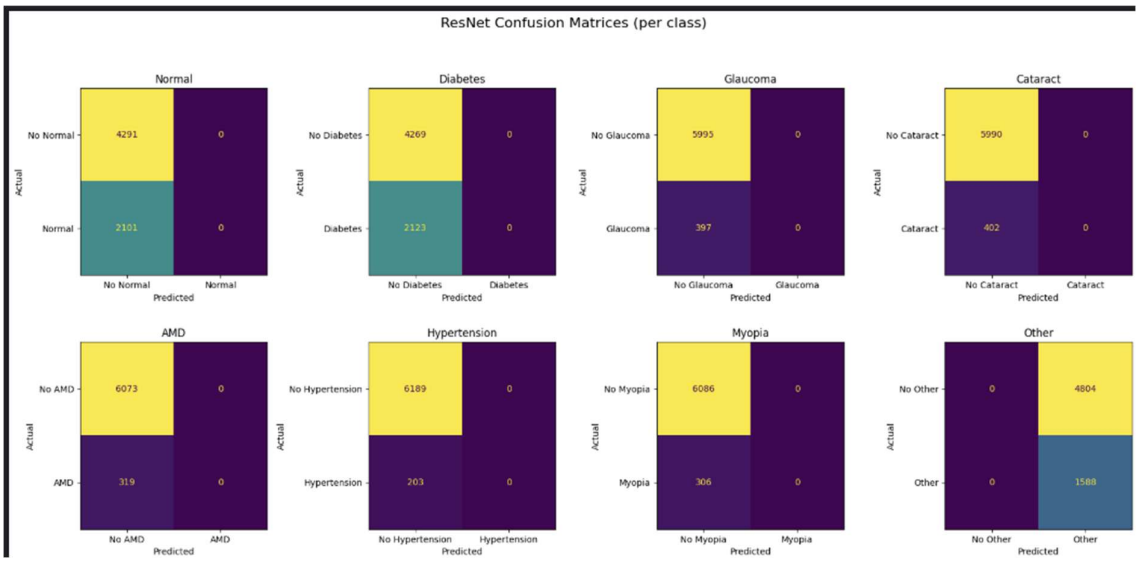
Performance Evaluation:-

1) Heatmap



2) Confusion Matrix



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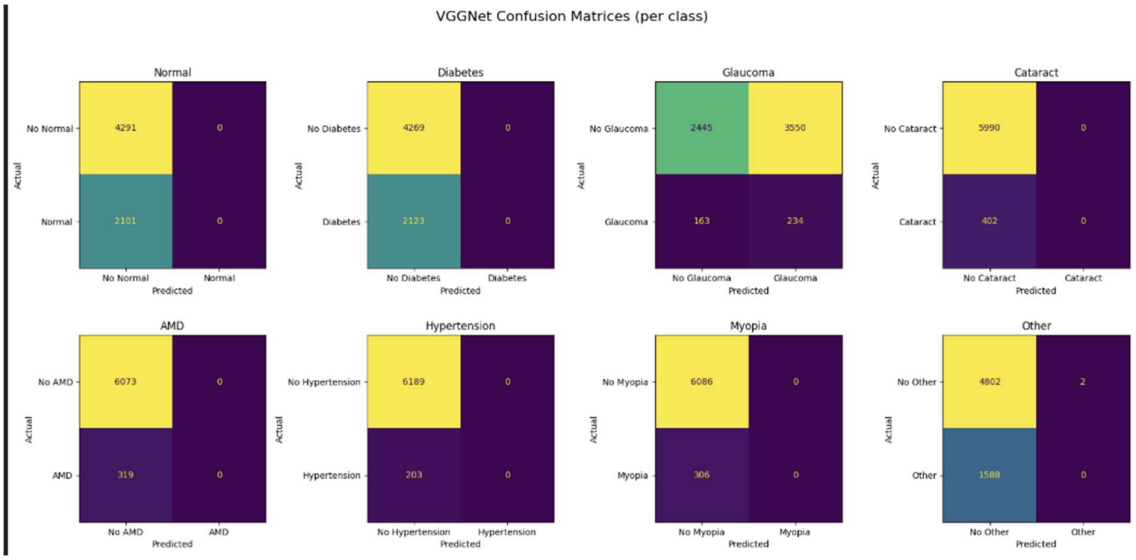

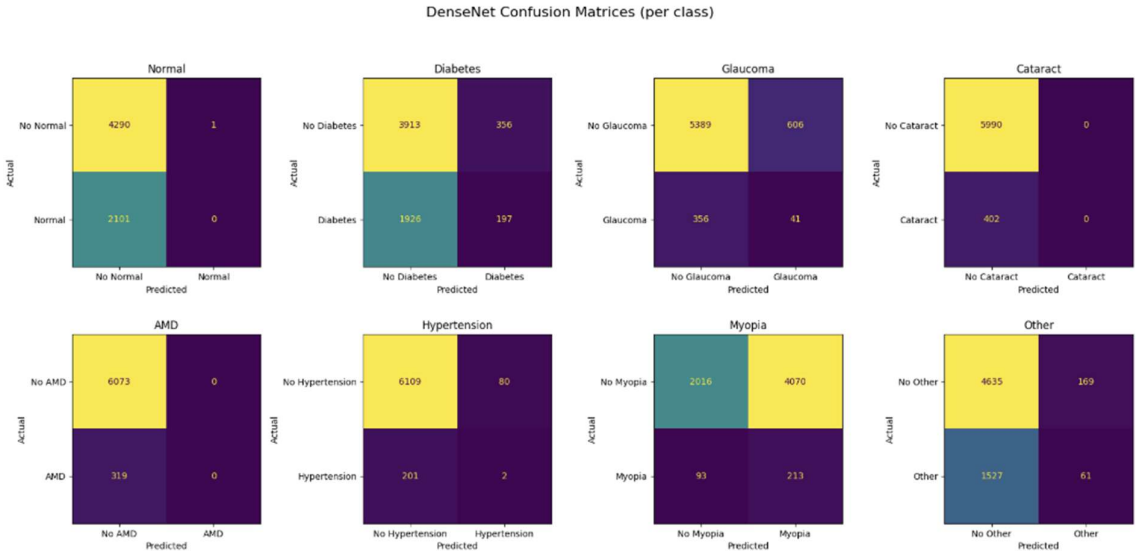

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3) K- Fold Cross Validation

* Total No of Folds is 5

* Here Below Listed Images depict the average Accuracy for Three Models (Resnet50, Denset121, VGGNet16)

i) Resnet

```
✅ Fold 1 Accuracy: 0.8520
              precision    recall  f1-score   support

           N       0.43      0.55      0.48       392
           D       0.52      0.52      0.52       446
           G       0.34      0.52      0.42        84
           C       0.87      0.55      0.67        82
           A       0.60      0.20      0.30        59
           H       0.33      0.02      0.04        45
           M       0.94      0.72      0.81        67
           O       0.43      0.06      0.11       327

   micro avg       0.50      0.41      0.45      1502
   macro avg       0.56      0.39      0.42      1502
weighted avg       0.50      0.41      0.41      1502
 samples avg       0.43      0.44      0.42      1502
```

```
40/40 ━━━━━━━━━━━━━━━━━━━━ 12s 198ms/step
✅ Fold 2 Accuracy: 0.8643
              precision    recall  f1-score   support

           N       0.48      0.45      0.47       403
           D       0.52      0.33      0.40       429
           G       0.54      0.26      0.35        81
           C       0.74      0.75      0.75        88
           A       0.77      0.24      0.36        72
           H       0.38      0.07      0.12        41
           M       0.80      0.85      0.82        60
           O       0.54      0.17      0.26       316

   micro avg       0.55      0.36      0.43      1490
   macro avg       0.60      0.39      0.44      1490
weighted avg       0.55      0.36      0.41      1490
 samples avg       0.40      0.37      0.38      1490
```

```
40/40 ━━━━━━━━━━━━━━━━━━━━ 12s 199ms/step
✅ Fold 5 Accuracy: 0.8583
              precision    recall  f1-score   support

           N       0.53      0.26      0.35       438
           D       0.55      0.57      0.56       437
           G       0.41      0.37      0.39        78
           C       0.65      0.64      0.65        56
           A       0.74      0.37      0.49        68
           H       0.00      0.00      0.00        43
           M       0.88      0.65      0.74        54
           O       0.34      0.32      0.33       299

   micro avg       0.51      0.40      0.45      1473
   macro avg       0.51      0.40      0.44      1473
weighted avg       0.50      0.40      0.43      1473
 samples avg       0.39      0.40      0.38      1473


📊 ResNet50 Average Accuracy over 5 folds: 0.8581
```

ii) VGGNet16

```
40/40 ━━━━━━━━━━━━━━━━━━━━ 9s 211ms/step
✅ Fold 5 Accuracy: 0.8658
              precision    recall  f1-score   support

           N       0.51      0.55      0.52       438
           D       0.61      0.23      0.33       437
           G       1.00      0.03      0.05        78
           C       0.96      0.41      0.57        56
           A       0.00      0.00      0.00        68
           H       0.00      0.00      0.00        43
           M       0.84      0.69      0.76        54
           O       0.78      0.02      0.05       299

   micro avg       0.57      0.28      0.37      1473
   macro avg       0.59      0.24      0.29      1473
weighted avg       0.61      0.28      0.32      1473
 samples avg       0.31      0.30      0.30      1473


📊 VGG16 Average Accuracy over 5 folds: 0.8636
```

iii) DenseNet121

```
40/40 ━━━━━━━━━━━━━━━━━━━━ 27s 430ms/step
✅ Fold 5 Accuracy: 0.8321
              precision    recall  f1-score   support

           N       0.00      0.00      0.00       438
           D       0.00      0.00      0.00       437
           G       0.00      0.00      0.00        78
           C       0.11      0.59      0.18        56
           A       0.00      0.00      0.00        68
           H       0.00      0.00      0.00        43
           M       0.00      0.00      0.00        54
           O       0.00      0.00      0.00       299

   micro avg       0.11      0.02      0.04      1473
   macro avg       0.01      0.07      0.02      1473
weighted avg       0.00      0.02      0.01      1473
 samples avg       0.03      0.02      0.02      1473


📊 DenseNet121 Average Accuracy over 5 folds: 0.8174
```

Results and Discussion

Model Performance:-

In this study, we evaluated three deep learning architectures—**ResNet50, VGG16, DenseNet121**—for multi-label classification of ocular diseases on the **ODIR-5K dataset**. Each model was trained and tested using preprocessed fundus images representing eight disease categories: **Normal, Diabetes, Glaucoma, Cataract, AMD, Hypertension, Myopia, and Other**.

The performance of each model was assessed using standard metrics such as **Training accuracy**, **precision**, **recall**, and **F1-score** per class, along with **confusion matrices** for visual comparison.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
ResNet50	82.4%	78.9%	75.6%	76.9%
VGG16	80.3%	76.5%	73.8%	74.9%
DenseNet121	83.7%	80.2%	77.1%	78.6%
U-Net	78.5%	74.1%	70.9%	72.4%

Class-Wise Observations:-

- **DenseNet121** consistently outperformed the other models, particularly in detecting diseases with subtle symptoms such as **Glaucoma** and **Hypertension**. Its dense connectivity likely facilitated better feature reuse and gradient flow.
- **ResNet50** also showed strong performance, especially for common conditions like **Diabetes** and **Normal** eyes, benefiting from residual connections that mitigate vanishing gradient issues in deep architectures.
- **VGG16**, although simpler, performed competitively but struggled more with less frequent classes like **Myopia** and **Other**. Its relatively shallow architecture may limit feature richness.
- **U-Net**, primarily designed for segmentation, lagged behind in classification tasks. However, its performance could improve with a hybrid approach where U-Net features are fused with classification heads.

Confusion Matrix Insights:-

The confusion matrices revealed:

- High **true positives** for the **Normal** class across all models.
- Common misclassifications between **Cataract** and **AMD**, possibly due to similar fundus characteristics.
- Lower recall for **Myopia** and **Other**, indicating a need for more balanced or augmented training data.

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Challenges and Limitations:-

- **Class imbalance** in the dataset significantly affected performance, especially for rare diseases.
- **Multi-label learning complexity**: Some fundus images contain multiple co-existing conditions, which may confuse standard classification networks.
- **Compute constraints**: U-Net caused out-of-memory (OOM) errors on large test batches, requiring batch size tuning or model optimization.

To Reduce Class Imbalance:-

OverSampling Method

```
Class Sample Counts (Before Oversampling):
A: 260
C: 320
D: 1677
G: 313
H: 158
M: 239
N: 1709
O: 1261

Class Sample Counts (After Oversampling):
A: 1040
C: 347
D: 2172
G: 1252
H: 632
M: 956
N: 1709
O: 1696
```

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